AWE NUCLEAR SECURITY TECHNOLOGIES

EPOCH Workshop 2026: Diagnostics, probes, subsets

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Data filtering techniques

- Big simulations have a lot of data! Keep it manageable
  - Probes take less memory than particle dumps
  - Distribution functions also better than particle dumps
  - Use subsets to minimise particles in particle dumps
  - Use multiple output blocks with different dt_snapshot
- Some vars support single-precision output, useful if binning data
- Some vars let you output values averaged over a few steps, to reduce noise
- See:

https://epochpic.github.io/documentation/input_deck/input_deck_output_block.html

```
begin:subset
   name = background
   random_fraction = 0.1
   include_species:electron
   include_species:proton
end:subset

begin:subset
   name = high_gamma
   gamma_min = 1.3
   include_species:electron
end:subset

begin:output
   particles = background + high_gamma + always
   px = background + high_gamma
   py = background
   pz = always
```

```
begin:output
   file_prefix = grid_vars
   name = regular_dump
   dt_snapshot = 10.0e-15
   ex = always + single
   ey = always + single
   ez = always + single
   bx = always + single
   by = always + single
   bz = always + single
   temperature = always + species + single
   poynt_flux = always + single
   number_density = always + species + single
end:output

begin:output
   file_prefix = particle_data
   name = particle_dump
   dt_snapshot = 100e-15
   px = always + single
   py = always + single
   particle_weight = always + single
end:output
```


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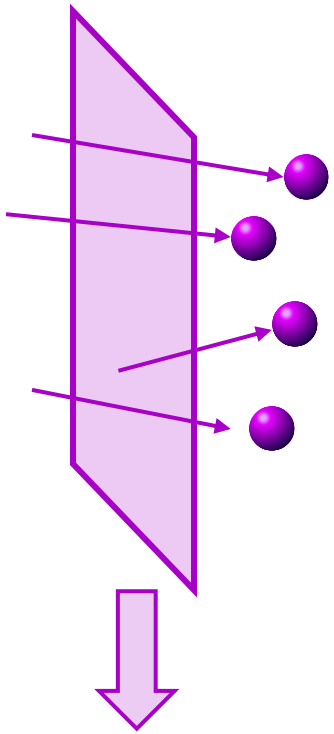
Particle probes

- Special type of output
  - Collects properties of particles which pass a “probe”
  - Defined by a point/line/plane in 1D/2D/3D simulations
  - Only saves particles passing in one direction (the normal)
  - Can filter by species type and particle energy
  - ek_max = -1.0 means no upper limit for kinetic energy

- Probes dump position, momentum and weight by default, when particles pass the probe
  - Optionally: particle ID, and probe-crossing-time too

- Probes only written to file when particles have passed the probe
  - Only writes particles which have passed since the last dump

- Useful if there are too many particles to dump otherwise



Output particle properties at crossing-time to file

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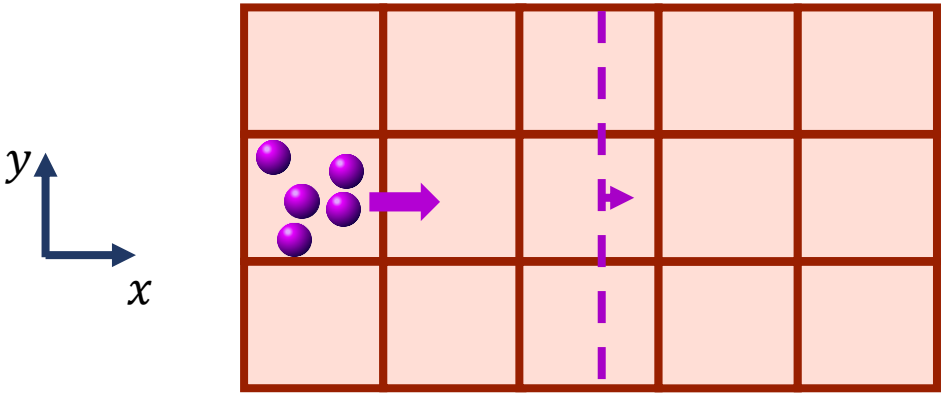
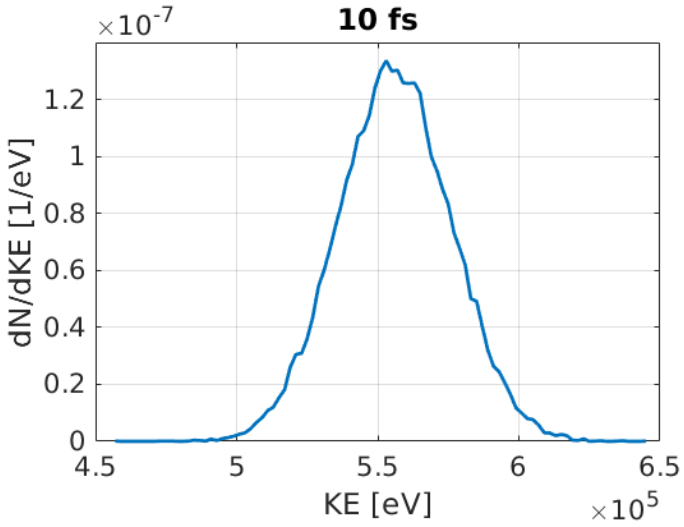
Example 5.1: Particle probes

- Plot the energy spectrum of a particle bunch when it passes a point in space
  - 2D simulation, e^- bunch drifts along window
  - Particle probe positioned at $x = 4\ \mu m$
  - Probe normal points in positive x, only logs particles moving from low-x to high-x.

```
begin:output
    dt_snapshot = t_end
    particle_probes = always
end:output

begin:probe
   name = Electron_probe
   point = (4.0e-6, 0.0)
   normal = (1.0, 0.0)
   ek_min = 0.0
   ek_max = -1.0
   include_species:Electron
   dumpmask = always
end:probe
```





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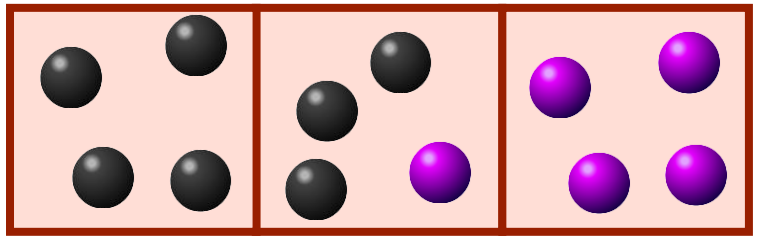
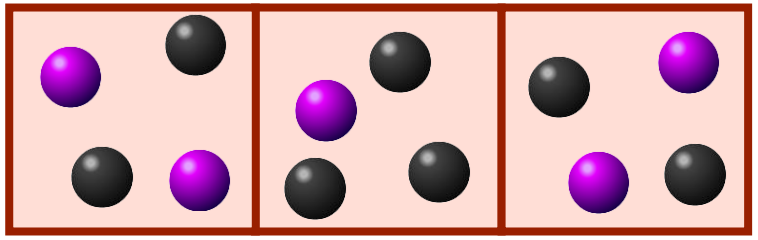
Subsets

- Particle dumps can be expensive
  - Subsets only plot specific particles
  - Can define by position (region of interest)
  - Can select random sample (spectrum estimate)
  - By momentum component
  - By total energy (specify min/max γ factor)

- Subsets can also be used to output field values over smaller cell-ranges
  - Useful for rapid outputs of small region



Position subset



Random subset

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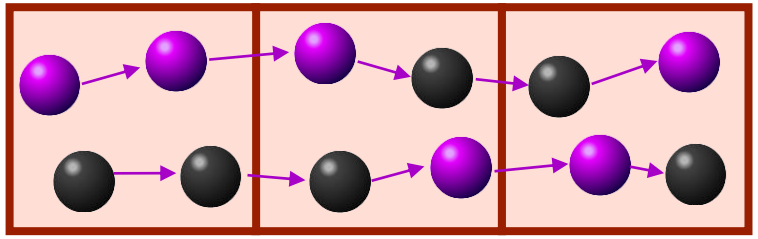
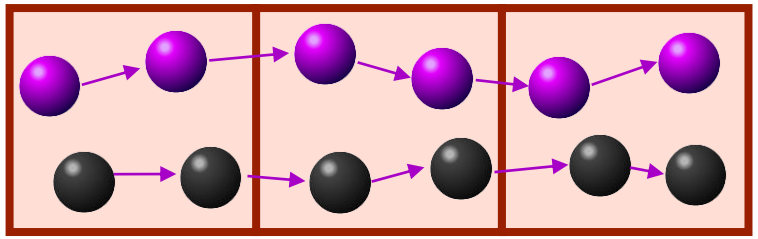
Persistent subsets

- By default, particles selected at time of dump
  - Subset particles can change during simulation
  - Persistent subsets follow specific particles

- Even in persistent subsets, dump-order is not fixed
  - E.g. first output in “momentum/px” will not always be same particle between dumps
  - To track specific particles, need to output ID



Default random subset



Persistent random subset

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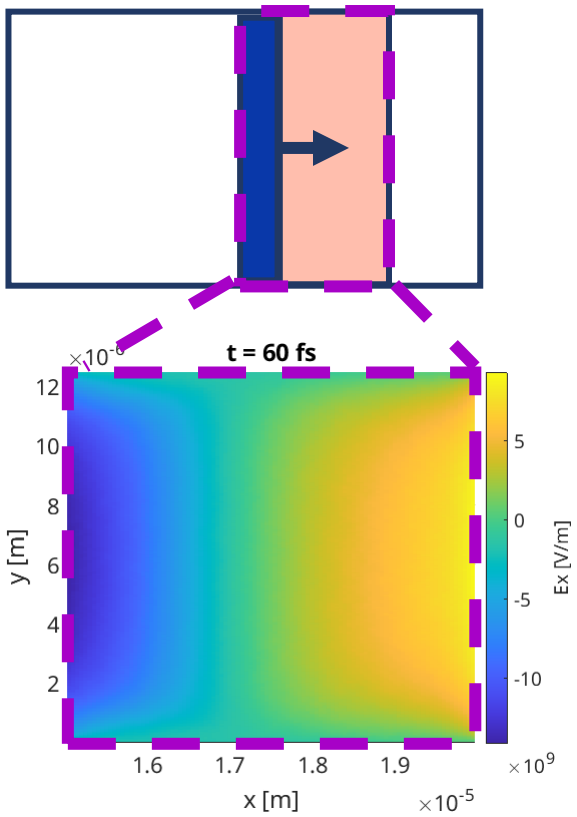
Example 5.2: Subsets

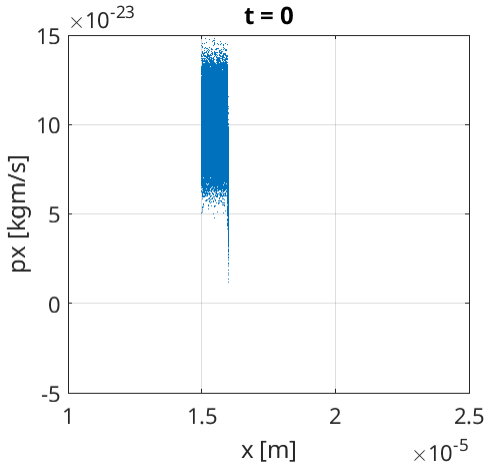
- Initialise neutral target, with boosted e^- on one side
  - Plot the (x, p_x) phase space of initially boosted e^- (position-based persistent subset)
  - Use macro-particle x and p_x, ignore weight
- Plot the Ex field in a domain subset around the target

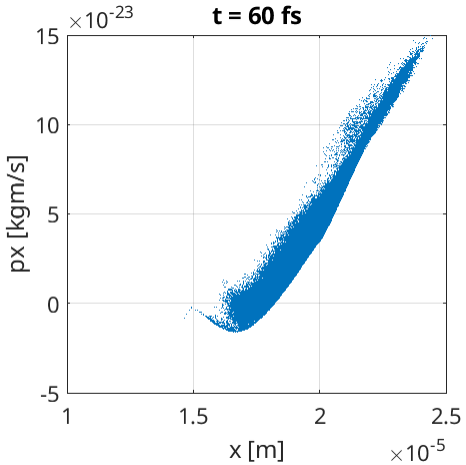
```
begin:subset
    name = Refluxers
    x_max = 16.0e-6
    persist_start_time = 0
    include_species:Electron
end:subset

begin:subset
   name = Target
   x_min = 15.0e-6
   x_max = 20.0e-6
end:subset

begin:output
    dt_snapshot = t_end / 10
    px = Refluxers
    ex = Target
    dt_average = 1.0e-15
end:output
```







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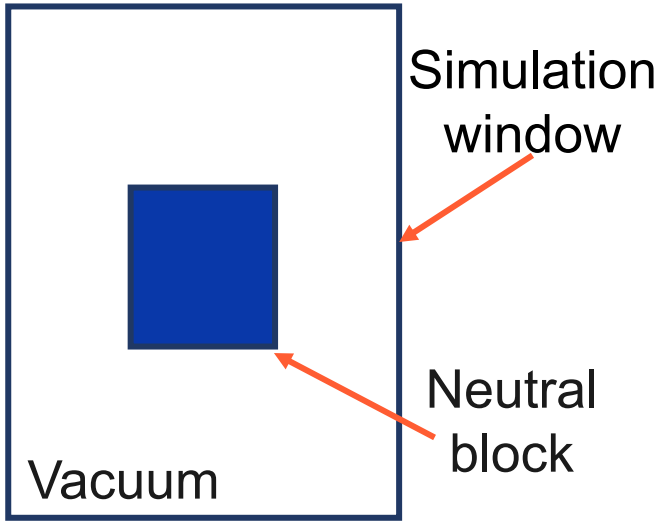
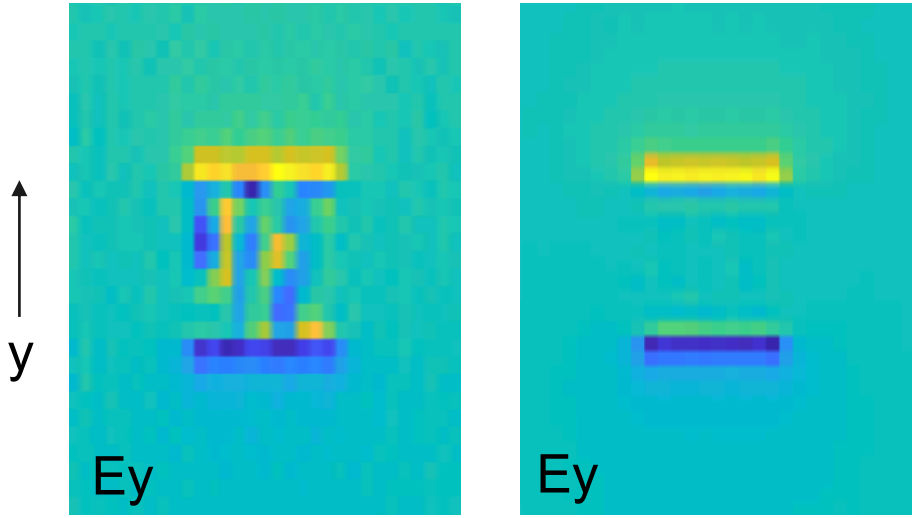
Output averaging



- PIC simulations are noisy
  - Large macro-particles moving: big currents, big fields
  - Local non-physical fluctuations can obscure physical fields
- Fluctuations are reduced when averaging fields in cells over many time-steps
- Example: sheath fields around target
  - Initialise finite neutral block with thermal electrons
  - Electrons move to escape, but fields grow to trap them
  - Internal fields: simulation noise
  - External fields: physical sheath



Raw snapshot Time average

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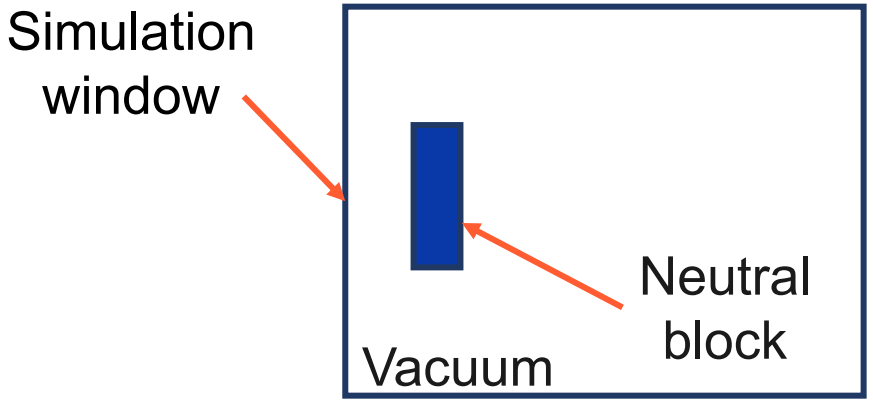
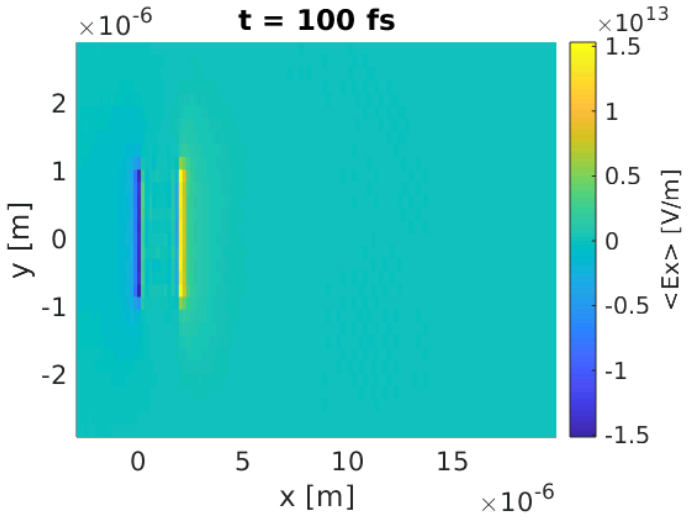
Example 5.3: Averaging

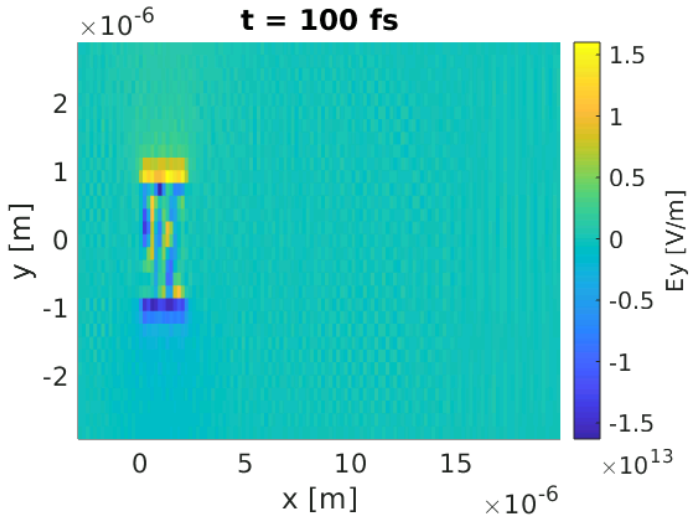
```
begin:output
  dt_snapshot = t_end / 5
  ey = always
  dt_average = 5.0e-15
  ex = always + average
end:output
```

- Model the sheath-field example
  - Take a snapshot of raw Ey fields
  - Also give time-averaged Ex fields

- Use a 5 fs averaging window, and output every 20 fs







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Restarting simulations from input deck

```
begin:control
  nx = 500
  ny = 500
  t_end = 100 * femto
  x_min = 0.0
  x_max = 10.0e-6
  y_min = -5.0e-6
  y_max = 5.0e-6
  stdout_frequency = 10

  restart_snapshot = 0003.sdf
end:control
```

- Previous demo: you write only relevant fields/properties
  - Restart dump: write EVERYTHING to file
- Restart dumps take a long time to write, and files are big
  - Only dump them when you need them
- Writing styles:
  - Dump at regular intervals
  - Dump at end of simulation
  - Dump at user request during run-time

```
begin:output
  name = normal
  file_prefix = normal
  dt_snapshot = 10.0e-15
  ey = always
end:output

begin:output
  name = restart
  file_prefix = restart
  restartable = T
  dt_snapshot = 20.0e-15
end:output
```


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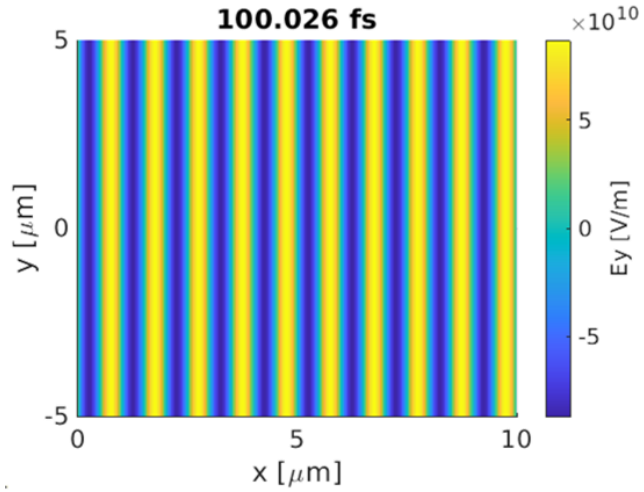
Example 5.4: Simulation restart

```
begin:control
  nx = 500
  ny = 500
  x_min = 0.0
  x_max = 10.0e-6
  y_min = -5.0e-6
  y_max = 5.0e-6
  stdout_frequency = 10

  t_end = 50 * femto        # Before restart
  # t_end = 100 * femto     # After restart

  # restart_snapshot = 5    # Uncomment to run from 0005.sdf
end:control
```

```
begin:output
  dt_snapshot = 10.0e-15
  ey = always
  force_final_to_be_restartable = T
end:output
```

- Simple laser input provided with commented lines
  - Run the simulation to 50 fs
  - Output block will make the last dump a restart dump
  - Next, uncomment the control-block lines
  - Simulation will restart at 50 fs, run to 100 fs

- Compare the resultant laser profile to a single 100 fs simulation, without restart
  - Are they the same?
  - How much larger is the restart SDF file?



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Restarting simulations with “STOP”

- When a file called STOP appears in the input.deck directory:
  - EPOCH prints a restart dump file
  - Simulation is then stopped, STOP file deleted
  - Useful if you can only run the code for a set amount of time

- If running this command before your EPOCH simulation:

  (**sleep** 60; touch STOP) &

  - A STOP file is created after 60 seconds
  - This is a good way to automatically trigger a restart-dump write
  - Good to add to Slurm job-scripts for cluster runs

- Try using a STOP file to make EPOCH write a restart-dump mid-job

```
touch STOP
```


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